

Version Control and Collaboration -- Guided Notes

Unit tp-4.4 | CompTIA Network+ N10-009 Obj. FC0-U71 4.4 | N10-008 FC0-U71 4.4

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Version control (also called source control or revision control) is a system that tracks _____ to code over time. It allows developers to _____ to any previous state, compare changes, and work _____ without overwriting each other's work.

Question 2

Git is a _____ version control system, meaning every developer has a _____ copy of the full repository history. SVN is a _____ version control system, meaning there is one central server. With Git, developers can _____ without internet access and sync later.

Question 3

A _____ is a snapshot of the repository at a point in time. When a developer saves changes, they create a commit with a _____ describing what changed. A _____ is a pointer to a specific commit and typically named for the development stream (e.g., main, develop, feature-login).

Question 4

A _____ request (also called a merge request) is a formal proposal to merge code from one branch into another. It triggers a _____ review where team members inspect the code before it is merged. This process catches bugs and enforces _____ standards before defects reach the main branch.

Question 5

A merge conflict occurs when two developers have edited the _____ lines of the same file in different branches. Git cannot automatically decide which version is correct, so it _____ the conflict and asks the developer to _____ resolve it. Frequent small commits reduce the severity of merge _____.

Question 6

GitHub, GitLab, and Bitbucket are _____ platforms that host Git repositories. They add features like _____ (tracks bugs and feature requests), _____ (automated testing pipelines), and team _____ (permissions, branch protection rules) on top of core Git.

Question 7

A .gitignore file tells Git to _____ tracking specific files or directories. Common entries include _____ (compiled output), _____ (local environment configs with secrets), and _____ (IDE configuration directories). Committing secrets to a public repo is a major _____ risk.

Question 8 -- Real World Application

REAL WORLD: A student accidentally commits their database password directly into their public GitHub repository in a config file. They realize it within 10 minutes and delete the file from the repo. Explain why deleting the file is NOT sufficient to protect the secret and what they must do to remediate the situation.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-4-4-version-control-and-collaboration/tp-4-4-version-control-and-collaboration-answer-key.pdf
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